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Chemistry Paper 3B WASSCE (SC), 2023

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Question 3B

(a) Determine the volume of water that should be added to 100 cm³ of 0.5 mol dm⁻³ HCl in order to obtain 0.3 mol dm⁻³ HCl. [5 marks]

(b) Describe briefly a chemical test to distinguish between dilute HCl and dilute HNO₃. Support the test with relevant equation. [6 marks]

Observation

Majority of the candidates responded to this question but their performance was below average.

In part (a), majority of the candidates could not solve this question.

In part (b), majority of the candidates could not describe a chemical test to distinguish between HCl and dilute HNO₃.

The expected answers include:

$$(a) C_1V_1 = C_2V_2$$

$$0.5 \times 100 = 0.3 \times V_2$$

$$V_2 = \frac{0.5 \times 100}{0.3}$$

$$= 166.67 \text{ cm}^3$$

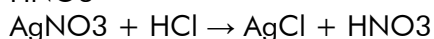
(accept 166.7 / 167)

$$\text{Volume of water} = 166.67 - 100$$

$$= 66.67 \text{ cm}^3$$

(b) Add AgNO₃(aq) to each of the solutions in a test tube. Formation of a white precipitate

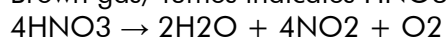
indicates that the solution is HCl, if no precipitate is formed then the solution is HNO₃



OR

Heat each sample in a test tube

Brown gas/ fumes indicates HNO_3 , no brown gas / fumes indicate HCl

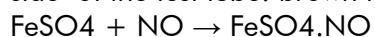


OR

To each sample in a test tube, add fresh $\text{FeSO}_4(\text{aq})$, followed by conc.

H_2SO_4 along the

side of the test tube: brown ring indicates HNO_3 , no brown ring indicates HCl



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